PBL Guideline

- Start with a problem statement
- Identify the portions of the problem that can be solved using image processing
- **Ground Rule**: The solution should incorporate atleast 3 of the following image processing techniques:
  1. *Image Enhancement*
  2. *Histogram Processing*
  3. *Spatial filtering*
  4. *Frequency domain filtering*
  5. *Noise removal*
  6. *Morphological Operations*
  7. *Color models*

BONUS points will be awarded to teams

1. Who go beyond standard RGB images (+1)
2. Compare results of 2 or more techniques for the given task (+1M)

1. Image Enhancement

Correcting exposure using intensity transformations

When to use: Image is washed out, too dark, or too flat.

TOOLKIT:

- **Contrast Stretching**: Expands range of intensity levels.
- **Gamma Correction**: Reveals details hidden in shadows/highlights.
- **Image Negative**: Inverts values to enhance white details in dark regions.

Real world Applications

- Making night-time road images clearer
- Improving readability of faded documents
- Enhancing low-light smartphone photos

Original (Dark) Gamma Corrected

2. Histogram Processing

Redistributes pixel intensity values to improve overall contrast.

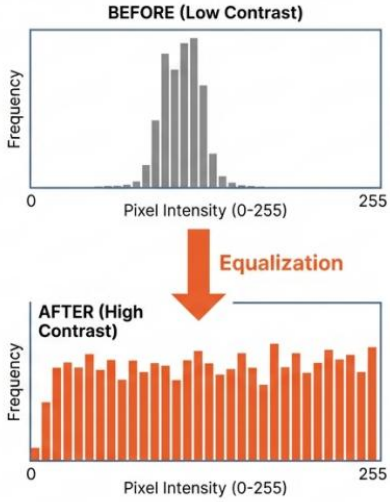
When to use: for images with poor contrast

Techniques:

- Histogram equalization
- Histogram matching
- CLAHE

Applications:

- Improving visibility in foggy images
- Enhancing medical or satellite images
- Standardizing images from different cameras



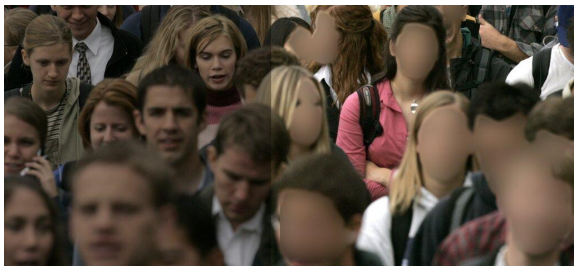
3. Spatial Filtering

Modifies each pixel using its neighboring pixels

When to use: If noise, blur, or edge clarity is a problem.

Techniques:

- Mean/median/gaussian filters
- Noise removal filters
- Edge detection filters



Applications:

- Cleaning noisy cctv footages
- Removing noise from medical images
- Detecting cracks or defects on surfaces

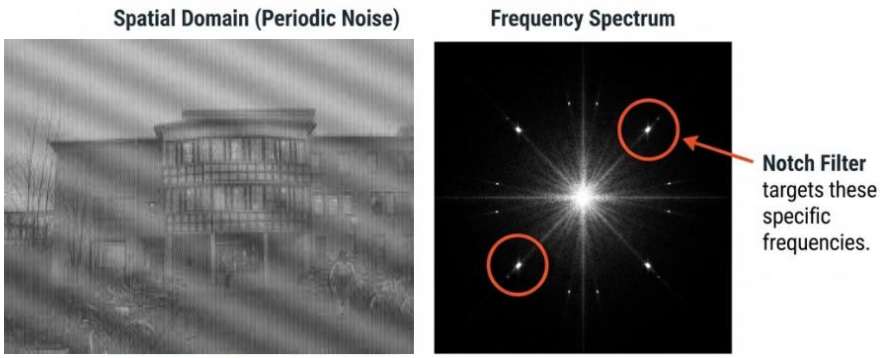
4. Frequency domain filtering

Processes images based on **low-frequency (smooth areas)** and **high-frequency (edges/details)** components.

When to use: If noise or blur has a **repetitive or structured pattern**..

Applications:

- Removing repetitive noise patterns
- Enhancing fine texture details
- Image compression preprocessing



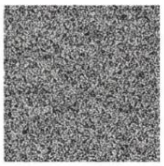
THE TOOLKIT:

- **Low-pass filters:** Attenuate high frequencies (Smoothing).
- **High-pass filters:** Attenuate low frequencies (Sharpening).
- **Band-pass / Notch filters:** The surgeon's tool—removing specific periodic noise without affecting the rest of the image.

5. Noise Modeling

Identifies the type of noise in an image and remove it using suitable methods.

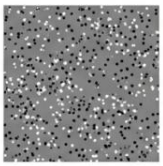
When to use: If the image is degraded due to **known physical causes**.



Gaussian Noise

Electronic sensor noise.

`Rx: Wiener Filtering` **(Minimizes mean square error).**



Salt-and-Pepper

Data transmission errors.

`Rx: Contra-harmonic Mean / Median Filter.`

Motion Blur

Camera or object movement.

`Rx: Inverse Filtering` **(Reversing known degradation).**

INSIGHT: One filter does not fit all. Restoration is about matching the filter to the noise source.

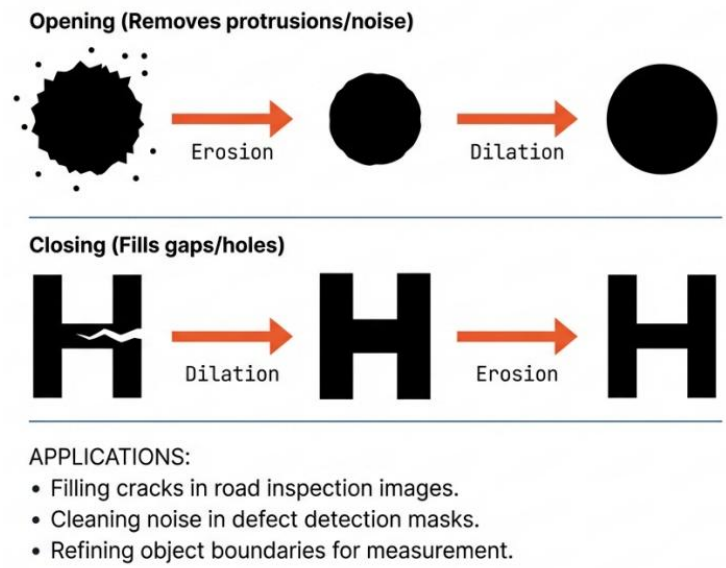
Applications:

- Restoring old photographs
- Cleaning images obtained from low cost sensors
- Removing motion blur

6. Morphological Operations

Works on the **shape and structure** of objects in binary or segmented images

When to use: Imperfect segmentation. Gaps in lines, small artifacts, or rough contours.



Toolkit:

- Erosion, Dilation, Opening, Closing, Hit-or-Miss

7. Color Space Transformations

Represents color in different ways to make processing easier.

When to use: When color information is crucial for the task to be performed

Toolkit:

- RGB, CMYK, LaB, HSV, etc

Applications:

- Fruit freshness detection
- Object sorting based on color
- Color-blind assistive tools
- Fabric or paint consistency checking
- Pseudo-color imaging

Problem Statement

Identify any real world problem that can be addressed with digital image processing.
The solution **should incorporate at least 3** of the following image processing techniques:

1. *Image Enhancement*
2. *Histogram Processing*
3. *Spatial filtering*
4. *Frequency domain filtering*
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Problem Statement

Rice is a staple food for millions of people, and its quality directly affects market value, consumer satisfaction, and export potential. ***Broken grains, discolored grains, foreign particles, and variations in size and appearance all contribute to perceived quality***. Farmers, traders, and rice mill owners routinely assess rice quality before pricing or distribution. Traditionally, this assessment is done by human inspection, where trained workers examine samples to check grain appearance, uniformity, and visible defects. However, this process is subjective and may vary from person to person.

Large volumes of rice are handled daily, especially during harvest seasons, making detailed manual inspection time-consuming and costly. Fatigue and inconsistency can lead to incorrect grading, which may cause disputes between sellers and buyers. Poor-quality rice may reach consumers, while good-quality produce may be undervalued. Small farmers, in particular, are disadvantaged when they lack access to standardized quality evaluation methods.

AIM Control
DOANH NGHIỆP TƯ NHÂ
TÂN PHÁT

Possible directions for Solutions

1. Improving Visibility of Grain Characteristics
2. Separating Individual Grains from Bulk Samples
3. Detecting Broken, Cracked, or Deformed Grains
4. Identifying discolored or aged grains
5. Identifying Foreign Particles and Impurities
6. Comparing Rice Samples Across Different Batches or Sources

Stepwise Problem-Solving Activities with Milestone (12 Weeks)

PA	ACTIVITY	MILESTONE
PA1	Identify problem to solve. Justify why traditional approaches fail to meet the requirements	M1 (Week 1)
PA2	Perform primary and secondary research	M2 (Week 2)
PA3	Finalize project requirements and plan project schedule **----------Generate Project Brief----------**	M3 (Week 3)
PA4	Reason about business modelling. Talk to clients about budget and expenses	M4 (Week 4)
PA5	Brainstorm possible solutions to the problem. Finalize solution. Identify image processing concepts relevant to the solution	M5 (Week 5-6)
PA6	Generate design draft. Identify components for the project. Assign responsibilities to team members	M6 (Week 7-8)
PA7	Implement individual components and perform unit testing	M7 (Week 9-10
PA8	Integrate components and test project. Get product verified by target customer	M8 (Week 11)
PA9	Demonstrate final project	End of week 12